

Numerical Method - Homework 1

B13902022 Yu-Chi,Lai

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I use the following code to reproduce the example ($f_i = 1, A = -64, B = 0$) given in the paper, and I got 0 using the first method ($\text{calc1}()$, $1 - p_i = 1 - \frac{1}{1+e^{Af_i+B}}$), while getting $1.60381e-28$ when using the second method ($\text{calc2}()$, $1 - p_i = \frac{e^{Af_i+B}}{1+e^{Af_i+B}}$)

The second result is more precise, this is mainly because, when $(f_i, A, B) = (1, -64, 0)$, $\exp(Af_i + B) \approx 0$, it successfully avoid the subtraction between two close numbers in calc1 (In this case, 1 and p_i), which leads to catastrophic cancellation.

```
1  #include <bits/stdc++.h>
2  using namespace std;
3
4  double calc1(double f, double A, double B) {
5      return 1.0 - 1.0 / (exp(A*f+B)+1.0);
6  }
7
8  double calc2(double f, double A, double B) {
9      return exp(A*f+B) / (1.0+exp(A*f+B));
10 }
11
12 int main() {
13     cout << calc1(1, -64, 0) << endl;
14     // output: 0
15     cout << calc2(1, -64, 0) << endl;
16     // output: 1.60381e-28
17 }
```
